

Outputs of SQL Query

1. Explore Dataset

View Scheme

```
PRAGMA table_info(sales);
```

	cid	name	type	notnull	dflt_value	pk
1	0	Row ID	INTEGER	0	<null>	0
2	1	Order ID	TEXT	0	<null>	0
3	2	Order Date	TEXT	0	<null>	0
4	3	Ship Date	TEXT	0	<null>	0
5	4	Ship Mode	TEXT	0	<null>	0
6	5	Customer ID	TEXT	0	<null>	0
7	6	Customer Name	TEXT	0	<null>	0
8	7	Segment	TEXT	0	<null>	0
9	8	Country	TEXT	0	<null>	0
10	9	City	TEXT	0	<null>	0
11	10	State	TEXT	0	<null>	0
12	11	Postal Code	INTEGER	0	<null>	0
13	12	Region	TEXT	0	<null>	0
14	13	Product ID	TEXT	0	<null>	0
15	14	Category	TEXT	0	<null>	0
16	15	Sub-Category	TEXT	0	<null>	0
17	16	Product Name	TEXT	0	<null>	0
18	17	Sales	REAL	0	<null>	0
19	18	Quantity	INTEGER	0	<null>	0
20	19	Discount	REAL	0	<null>	0
21	20	Profit	REAL	0	<null>	0

View Sample Records

```
SELECT * FROM sales LIMIT 10;
```

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City
1	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henders
2	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henders
3	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Ang
4	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort La
5	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort La
6	6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Ang
7	7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Ang
8	8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Ang
9	9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Ang
10	10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Ang

Total Records

```
SELECT COUNT(*) AS total_rows FROM sales;
```

	total_rows
1	9994

2. WHERE Filters

Orders from South Region

```
SELECT "Row ID", "Customer ID", "Customer Name",Region FROM sales WHERE Region = 'South';
```

	"Row ID"	"Customer ID"	"Customer Name"	Region
1	1	CG-12520	Claire Gute	South
2	2	CG-12520	Claire Gute	South
4	4	SO-20335	Sean O'Donnell	South
4	5	SO-20335	Sean O'Donnell	South
5	13	AA-10480	Andrew Allen	South
6	44	ES-14080	Erin Smith	South
7	70	KD-16270	Karen Daniels	South
8	73	JE-15745	Joel Eaton	South

Furniture Category Sales

```
SELECT "Row ID", "Customer ID", "Customer Name",Region, Category FROM sales WHERE Category = 'Furniture';
```

	"Row ID"	"Customer ID"	"Customer Name"	Region	Category
1	1	CG-12520	Claire Gute	South	Furniture
2	2	CG-12520	Claire Gute	South	Furniture
3	4	SO-20335	Sean O'Donnell	South	Furniture
4	6	BH-11710	Brosina Hoffman	West	Furniture
5	11	BH-11710	Brosina Hoffman	West	Furniture
6	24	SF-20065	Sandra Flanagan	East	Furniture
7	25	EB-13870	Emily Burns	West	Furniture
8	28	TB-21520	Tracy Blumstein	East	Furniture

Sales Greater Than 500

```
SELECT "Row ID", "Customer ID", "Customer Name",Region, Category, Sales FROM sales WHERE Sales > 500;
```

	"Row ID"	"Customer ID"	"Customer Name"	Region	Category	Sales
1	2	CG-12520	Claire Gute	South	Furniture	731.94
2	4	SO-20335	Sean O'Donnell	South	Furniture	957.5775
3	8	BH-11710	Brosina Hoffman	West	Technology	907.152
4	11	BH-11710	Brosina Hoffman	West	Furniture	1706.184
5	12	BH-11710	Brosina Hoffman	West	Technology	911.424
6	17	PK-19075	Pete Kriz	Central	Office Supplies	665.88
7	25	EB-13870	Emily Burns	West	Furniture	1044.63
8	28	TB-21520	Tracy Blumstein	East	Furniture	3083.43

Orders in 2017

```
SELECT "Row ID", "Customer ID", "Customer Name", Region, Category, "Order Date" FROM sales WHERE "Order Date" LIKE '%2017';
```

	Row ID	Customer ID	Customer Name	Region	Category	Order Date
1	13	AA-10480	Andrew Allen	South	Office Supplies	4/15/2017
2	24	SF-20065	Sandra Flanagan	East	Furniture	7/16/2017
3	35	MA-17560	Matt Abelman	Central	Office Supplies	10/19/2017
4	42	LC-16930	Linda Cazamias	Central	Technology	9/10/2017
5	44	ES-14080	Erin Smith	South	Office Supplies	9/19/2017
6	72	TB-21520	Tracy Blumstein	Central	Office Supplies	9/14/2017
7	76	KB-16600	Ken Brennan	Central	Office Supplies	12/9/2017
8	77	KB-16600	Ken Brennan	Central	Furniture	12/9/2017

3. GROUP BY Aggregations

Total Sales by Region

```
SELECT Region, ROUND(SUM(Sales),2) AS Total_Sales FROM sales GROUP BY Region ORDER BY Total_Sales DESC;
```

	Region	Total_Sales
1	West	725457.82
2	East	678781.24
3	Central	501239.89
4	South	391721.91

Total Profit by Region

```
SELECT Region, ROUND(SUM(Profit),2) AS Total_Profit FROM sales GROUP BY Region ORDER BY Total_Profit DESC;
```

	Region	Total_Profit
1	West	108418.45
2	East	91522.78
3	South	46749.43
4	Central	39706.36

Average Sales by Category

```
SELECT Category, ROUND(AVG(Sales),2) AS Avg_Sales FROM sales GROUP BY Category;
```

	Category	Avg_Sales
1	Furniture	349.83
2	Office Supplies	119.32
3	Technology	452.71

Total Quantity Sold by Category

```
SELECT Category, SUM(Quantity) AS Total_Quantity FROM sales GROUP BY Category;
```

	Category	Total_Quantity
1	Furniture	8028
2	Office Supplies	22906
3	Technology	6939

4. Sorting and Limiting

Top 10 Products by Sales

```
SELECT  
    "Product Name",  
    ROUND(SUM(Sales),2) AS Total_Sales  
FROM sales GROUP BY "Product Name" ORDER BY Total_Sales DESC LIMIT 10;
```

	"Product Name"	Total_Sales
1	Canon imageCLASS 2200 Advanced Copier	61599.82
2	Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	27453.38
3	Cisco TelePresence System EX90 Videoconferencing Unit	22638.48
4	HON 5400 Series Task Chairs for Big and Tall	21870.58
5	GBC DocuBind TL300 Electric Binding System	19823.48
6	GBC Ibimaster 500 Manual ProClick Binding System	19024.5
7	Hewlett Packard LaserJet 3310 Copier	18839.69
8	HP Designjet T520 Inkjet Large Format Printer - 24" Color	18374.9

Top 5 Customers

```
SELECT  
    "Customer Name",  
    ROUND(SUM(Sales),2) AS Total_Sales  
FROM sales GROUP BY "Customer Name" ORDER BY Total_Sales DESC LIMIT 5;
```

	"Customer Name"	Total_Sales
1	Sean Miller	25043.05
2	Tamara Chand	19052.22
3	Raymond Buch	15117.34
4	Tom Ashbrook	14595.62
5	Adrian Barton	14473.57

Top Categories

```
SELECT
    Category,
    ROUND(SUM(Sales),2) AS Sales
FROM sales GROUP BY Category ORDER BY Sales DESC;
```

	Category	Sales
1	Technology	836154.03
2	Furniture	741999.8
3	Office Supplies	719047.03

5. Business Use Cases

Monthly Sales Trend

```
SELECT
    substr("Order Date", -4) AS Year,
    CAST(substr("Order Date", 1, instr("Order Date", '/') - 1) AS INTEGER) AS Month,
    ROUND(SUM(Sales), 2) AS Total_Sales
FROM sales GROUP BY Year, Month ORDER BY CAST(Year AS INTEGER), Month;
```

	Year	Month	Total_Sales
1	2014	1	14236.9
2	2014	2	4519.89
3	2014	3	55691.01
4	2014	4	28295.35
5	2014	5	23648.29
6	2014	6	34595.13
7	2014	7	33946.39
8	2014	8	27909.47

48 rows

Most Profitable Products

```
SELECT
    "Product Name",
    ROUND(SUM(Profit),2) AS Total_Profit
FROM sales GROUP BY "Product Name" ORDER BY Total_Profit DESC LIMIT 10;
```

	"Product Name"	Total_Profit
1	Canon imageCLASS 2200 Advanced Copier	25199.93
2	Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind	7753.04
3	Hewlett Packard LaserJet 3310 Copier	6983.88
4	Canon PC1060 Personal Laser Copier	4570.93
5	HP Designjet T520 Inkjet Large Format Printer - 24" Color	4094.98
6	Ativa V4110MDD Micro-Cut Shredder	3772.95
7	3D Systems Cube Printer, 2nd Generation, Magenta	3717.97
8	Plantronics Savi W720 Multi-Device Wireless Headset System	3696.28

10 rows

Least Profitable Products

```
SELECT
    "Product Name",
    ROUND(SUM(Profit),2) AS Total_Profit
FROM sales GROUP BY "Product Name" ORDER BY Total_Profit ASC LIMIT 10;
```

	"Product Name"	Total_Profit
1	Cubify CubeX 3D Printer Double Head Print	-8879.97
2	Lexmark MX611dhe Monochrome Laser Printer	-4589.97
3	Cubify CubeX 3D Printer Triple Head Print	-3839.99
4	Chromcraft Bull-Nose Wood Oval Conference Tables & Bases	-2876.12
5	Bush Advantage Collection Racetrack Conference Table	-1934.4
6	GBC DocuBind P400 Electric Binding System	-1878.17
7	Cisco TelePresence System EX90 Videoconferencing Unit	-1811.08
8	Martin Yale Chadless Opener Electric Letter Opener	-1299.18

10 rows ▾ | ↺ ⌚

Top Customers by Profit

```
SELECT
    "Customer Name",
    ROUND(SUM(Profit),2) AS Total_Profit
FROM sales GROUP BY "Customer Name" ORDER BY Total_Profit DESC LIMIT 10;
```

	"Customer Name"	Total_Profit
1	Tamara Chand	8981.32
2	Raymond Buch	6976.1
3	Sanjit Chand	5757.41
4	Hunter Lopez	5622.43
5	Adrian Barton	5444.81
6	Tom Ashbrook	4703.79
7	Christopher Martinez	3899.89
8	Keith Dawkins	3038.63

10 rows ▾ | ↺ ⌚

Region-wise Sales & Profit

SELECT

Region,

ROUND(SUM(Sales),2) AS Sales,

ROUND(SUM(Profit),2) AS Profit

FROM sales GROUP BY Region ORDER BY Sales DESC;

	Region	Sales	Profit
1	West	725457.82	108418.45
2	East	678781.24	91522.78
3	Central	501239.89	39706.36
4	South	391721.91	46749.43

4 rows

6. Duplicate Records

Check Duplicate Order IDs

SELECT

"Order ID",

COUNT(*) AS occurrences

FROM sales GROUP BY "Order ID" HAVING COUNT(*) > 1;

	"Order ID"	occurrences
1	CA-2014-100090	2
2	CA-2014-100363	2
3	CA-2014-100678	4
4	CA-2014-100706	2
5	CA-2014-100762	4
6	CA-2014-100895	3
7	CA-2014-100916	3
8	CA-2014-101560	4

1-500 of 501+

Exact Duplicate Rows

SELECT "Row ID", "Order ID", "Customer Name", "Product ID", Sales, COUNT(*) AS duplicate_count FROM sales

GROUP BY

"Order ID",

"Product ID",

Sales

HAVING COUNT(*) > 1;

	"Row ID"	"Order ID"	"Customer Name"	"Product ID"	Sales	duplicate_count
1	3406	US-2014-150119	Laurel Beltran	FUR-CH-10002965	281.372	2

7. Data Validation

Null Values

```
SELECT
    COUNT(*) AS Total_Rows
FROM sales WHERE "Order ID" IS NULL OR Sales IS NULL OR Profit IS NULL;
```

	Total_Rows	
1	0	

1 row ▾  

Distinct Customers

```
SELECT COUNT(DISTINCT "Customer ID") AS Customers FROM sales;
```

	Customers	
1	793	

Distinct Products

```
SELECT COUNT(DISTINCT "Product ID") AS Products FROM sales;
```

	Products	
1	1862	

1 row ▾  

Sales Validation

```
SELECT ROUND(SUM(Sales),2) AS Total_Sales FROM sales;
```

	Total_Sales	
1	2297200.86	

1 row ▾  